ReLoop – Online Marketplace for Refurbished Devices

E-Commerce Web Application

Project Documentation

Website Name:Classico

Topics Covered: Tech Backbone, Payment Solutions, Security

Problem Statement

Develop a fully functional online store (e.g., bookstore or electronics) that allows product browsing, cart management, secure checkout, and dummy payment integration.

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1. Introduction

E-commerce, also known as electronic commerce, refers to the buying and selling of goods and services over the internet. With the rapid growth of technology and the widespread use of smartphones, e-commerce has become an essential part of modern business, reshaping how consumers shop and how businesses operate.This project - **Online Marketplace for Refurbished Devices** - focuses on developing a fully functional e-commerce web application that allows users to browse a product catalog, add items to a shopping cart, provide shipping details, and complete purchases through a secure online payment gateway. It provides a convenient and efficient platform for both customers and sellers.

2. Technologies Used

The development of **Online Marketplace for Refurbished Devices** requires a combination of technologies spanning user interface design, server-side logic, data management, and payment processing. The stack follows a three-tier model: frontend, backend, and database.

Category	Technology	Purpose in Project
Frontend	JavaScript (ES6+)	Page structure and semantic markup for all views
Backend	Node.js	Styling, layout, animations, and responsive design
Frontend	JavaScript (ES6+)	Cart logic, DOM manipulation, and Fetch API calls
Backend	Node.js	Server-side JavaScript runtime environment
Backend	Express.js	REST API routing, middleware, and request handling
Database	MySQL	Stores products, orders, order items, and customer details
Payment	PayPal JS SDK	Smart Payment Buttons for PayPal and card checkout
Config	dotenv (npm)	Manages API credentials and DB config via .env file
DB Driver	mysql2 (npm)	Node.js to MySQL connection and query execution

3. Project Architecture

Tier	Layer Name	Technology	Responsibility
1	Presentation Layer	HTML / CSS / JavaScript	Renders the UI; manages cart state; calls the REST API
2	Application Layer	Node.js + Express.js	Processes requests; enforces business logic; calls PayPal API
3	Data Layer	MySQL (ecom_db)	Persists products, orders, and order line items

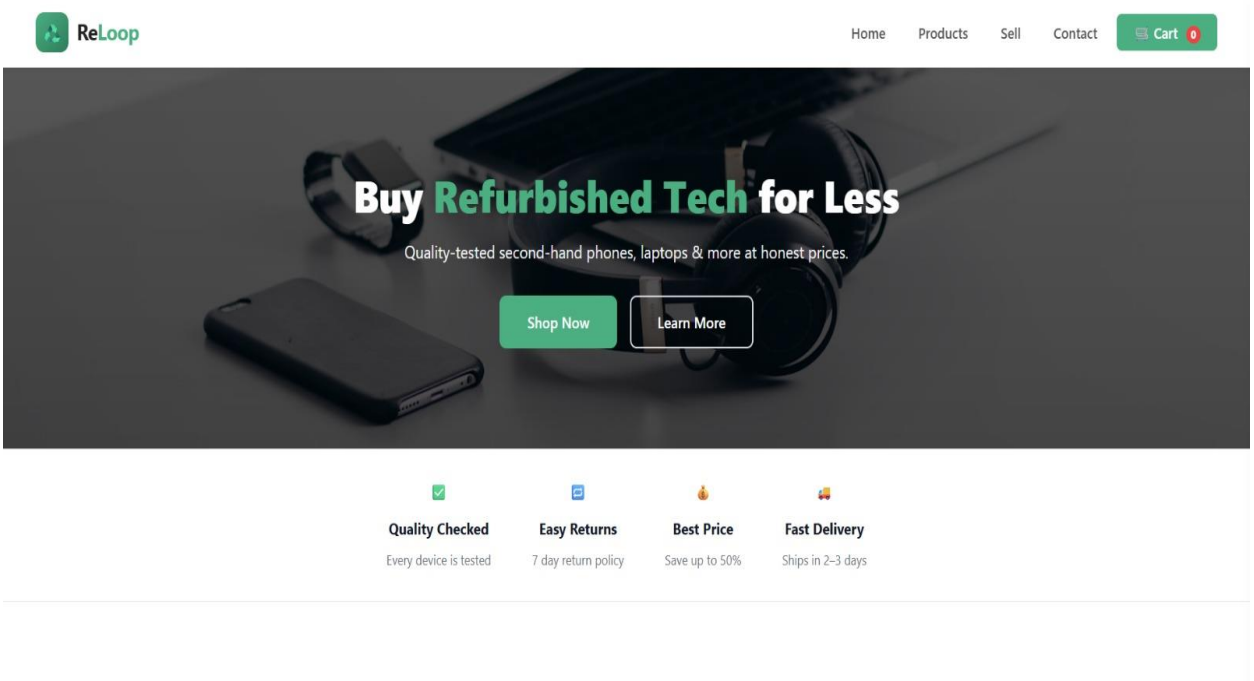
4. Challenges & Considerations

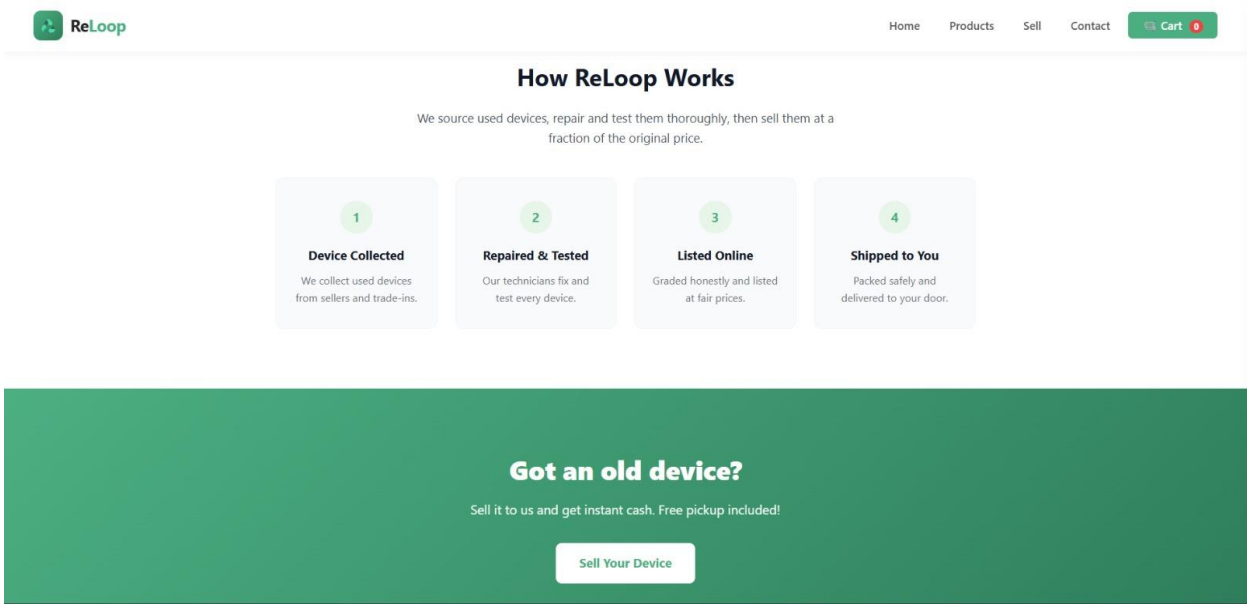
- **Security -** Handling customer data and payment information requires HTTPS, input sanitisation, and secure credential management (addressed via PayPal's PCI-compliant SDK and dotenv).
- **No Physical Interaction -** Customers cannot inspect products physically, making high-quality images and accurate descriptions critical.
- **Internet Dependency -** Both customers and the business rely on stable internet connectivity for the platform to function.
- **Scalability -** As the product catalog and order volume

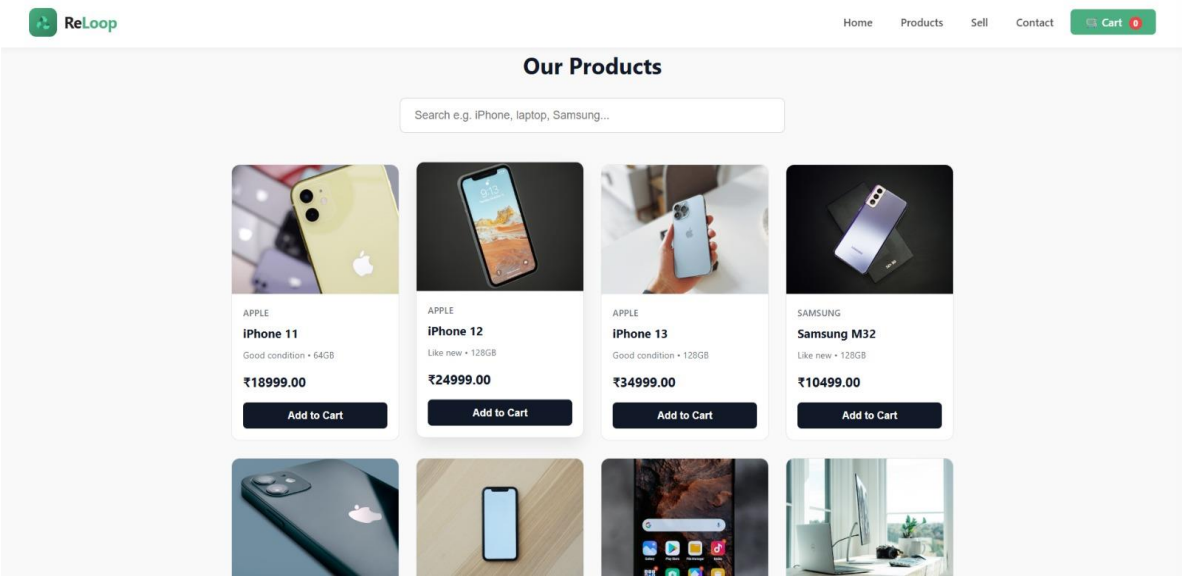
5. Conclusion

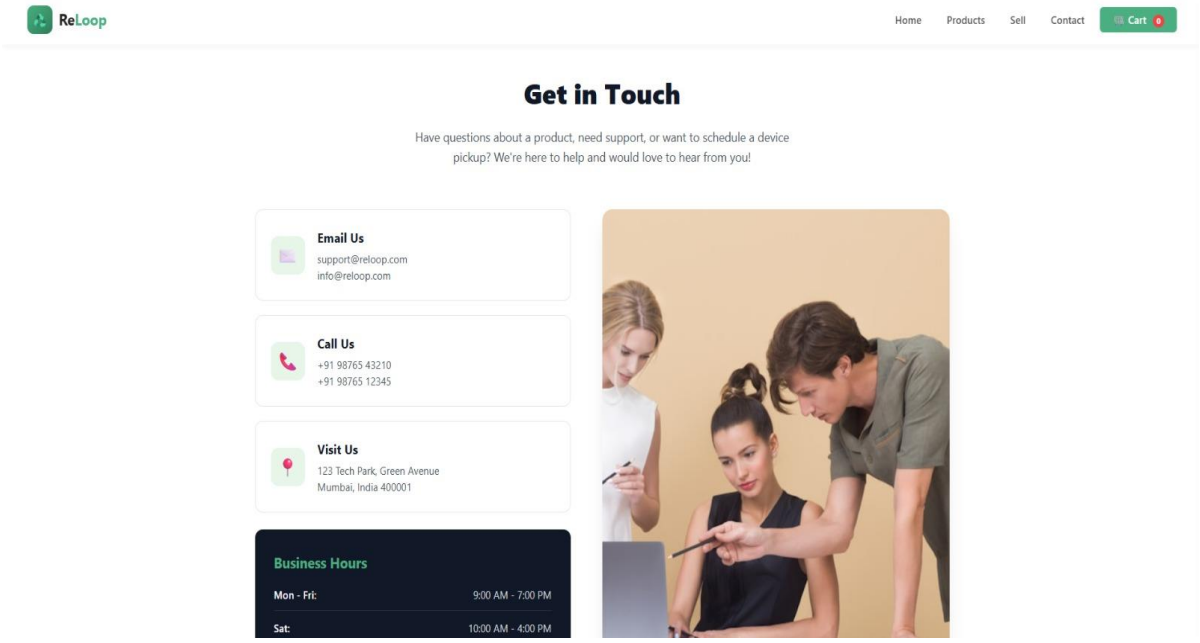
Online Marketplace for Refurbished Devices successfully demonstrates how a complete, functional e-commerce workflow can be built from scratch using fundamental, industry-standard web technologies. By deliberately avoiding heavy frontend frameworks and relying on a minimal Express backend, every layer of the application remains transparent, well-understood, and easy to extend.

Frontend :


ReLoop
Home
Products
Sell
Contact
Cart 0
Buy Refurbished Tech for Less
Quality-tested second-hand phones, laptops & more at honest prices.
Shop Now
Learn More
Quality Checked
Every device is tested
Easy Returns
7 day return policy
Best Price
Save up to 50%
Fast Delivery
Ships in 2–3 days


ReLoop
Home
Products
Sell
Contact
Cart 0
How ReLoop Works
We source used devices, repair and test them thoroughly, then sell them at a fraction of the original price.
1
Device Collected
We collect used devices from sellers and trade-ins.
2
Repaired & Tested
Our technicians fix and test every device.
3
Listed Online
Graded honestly and listed at fair prices.
4
Shipped to You
Packed safely and delivered to your door.
Got an old device?
Sell it to us and get instant cash. Free pickup included!
Sell Your Device


ReLoop
Home
Products
Sell
Contact
Cart 0
Our Products
Search e.g. iPhone, laptop, Samsung...
APPLE
iPhone 11
Good condition • 64GB
₹18999.00
Add to Cart
APPLE
iPhone 12
Like new • 128GB
₹24999.00
Add to Cart
APPLE
iPhone 13
Good condition • 128GB
₹34999.00
Add to Cart
SAMSUNG
Samsung M32
Like new • 128GB
₹10499.00
Add to Cart


ReLoop
Home
Products
Sell
Contact
Cart 0
Get in Touch
Have questions about a product, need support, or want to schedule a device pickup? We're here to help and would love to hear from you!
Email Us
support@reloop.com
info@reloop.com
Call Us
+91 98765 43210
+91 98765 12345
Visit Us
123 Tech Park, Green Avenue
Mumbai, India 400001
Business Hours
Mon - Fri: 9:00 AM - 7:00 PM
Sat: 10:00 AM - 4:00 PM

ReLoop

Home Products Sell Contact

Shipping Address

Full Name
testing-shipping

Email Address
shipping@sclkc.com

Address
testing-shipping-address

City
pune

State / Province
maharastra

Zip / Postal Code
1230342

Country
india

Order Summary

1x MacBook Air M1	₹55000
1x Mi Band 6	₹1499
Subtotal	₹56499
Shipping	Free
Total	**₹56499**

PayPal

Debit or Credit Card

Powered by PayPal

127.0.0.1:5500/checkout.html

sandbox.paypal.com/pay/checkout?atomic-event-state=eyJkb21haW4iOiJzZ...

JD PayPal

Pay in full
$706.00 today

Pay in 4
4 interest-free payments of $176.50. No late fees.
Terms

Pay Monthly
Plans as low as $38.04/mo for 24 months.
Terms

Pay with

Split this payment

CREDIT UNION 1 (AK)
Checking ••4440
$706.00

Visa
Credit ••9973

Show all payment methods

PayPal balance
$5,000.00 available
Unavailable at this time

Add card

Pay $706.00

About payment methods

www.sandbox.paypal.com/.../pr...

Backend & Database :

```
MySQL localhost:33060+ ssl reloop_db SQL > select * from products;
```

id	brand	name	condition_desc	attributes	price	image_url
1	Apple	iPhone 11	Good condition	64GB	18999.00	https://images.unsplash.com/photo-1574755393849-623942496936?w=400&h=300&fit=crop
2	Apple	iPhone 12	Like new	128GB	24999.00	https://images.unsplash.com/photo-1605236453806-6ff36851218e?w=400&h=300&fit=crop
3	Apple	iPhone 13	Good condition	128GB	34999.00	https://images.unsplash.com/photo-1632661674596-df8be070a5c5?w=400&h=300&fit=crop
4	Samsung	Samsung M32	Like new	128GB	10499.00	https://images.unsplash.com/photo-1610945265064-0e34e5519bbf?w=400&h=300&fit=crop
5	Samsung	Samsung S21	Good condition	256GB	27000.00	https://images.unsplash.com/photo-1616348436168-de43ad0db179?w=400&h=300&fit=crop
6	iPhone	iPhone 9	Good condition	128GB	16500.00	https://images.unsplash.com/photo-1585060544812-6b45742d762f?w=400&h=300&fit=crop
7	Xiaomi	Redmi Note 10	Like new	64GB	7999.00	https://images.unsplash.com/photo-1598327105666-5b89351aff97?w=400&h=300&fit=crop
8	Dell	Dell Inspiron 15	i5	8GB RAM · 256GB	22500.00	https://images.unsplash.com/photo-1593642632559-0c6d3fc62b89?w=400&h=300&fit=crop
9	HP	HP Pavilion 15	i5	8GB RAM · 512GB	26000.00	https://images.unsplash.com/photo-1496181133206-80ce9b88a853?w=400&h=300&fit=crop
10	Lenovo	Lenovo ThinkPad E14	i5	16GB RAM · 512GB	30000.00	https://images.unsplash.com/photo-1525547719571-a2d4ac8945e2?w=400&h=300&fit=crop
11	Apple	MacBook Air M1	Good condition	256GB	55000.00	https://images.unsplash.com/photo-1517336714731-489689fd1ca8?w=400&h=300&fit=crop
12	Asus	Asus VivoBook 15	i3	8GB RAM · 256GB	18000.00	https://images.unsplash.com/photo-1541807084-5c52b6b3adef?w=400&h=300&fit=crop
13	Sony	Sony WH-CH510	Good condition	Wireless	1899.00	https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=400&h=300&fit=crop
14	JBL	JBL Tune 230 TWS	Good condition	Wireless	2200.00	https://images.unsplash.com/photo-1572536147248-ac59a8abfa4b?w=400&h=300&fit=crop
15	Apple	Apple AirPods 2	Good condition	Wireless	7500.00	/02yGOcE9XNRd6kusx447Pov-item1.fit_lim.size_640x358.v1725983147.webp
16	Apple	Apple Watch Series 6	Good condition	44mm	18000.00	https://images.unsplash.com/photo-1546868871-7041f2a55e12?w=400&h=300&fit=crop
17	Samsung	Samsung Galaxy Watch 4	Like new	40mm	9500.00	https://images.unsplash.com/photo-1508685096489-7aacd43bd3b1?w=400&h=300&fit=crop
18	Xiaomi	Mi Band 6	Like new	Fitness band	1499.00	https://images.unsplash.com/photo-1575311373937-040b8e1fd5b6?w=400&h=300&fit=crop
19	Apple	iPad 9th Gen	Good condition	64GB WiFi	22000.00	https://images.unsplash.com/photo-1544244015-0df4b3ffc6b0?w=400&h=300&fit=crop
20	Samsung	Samsung Tab A8	Like new	64GB WiFi	12500.00	https://images.unsplash.com/photo-1561154464-82e9adf32764?w=400&h=300&fit=crop
21	Canon	Canon 1500D DSLR	Good condition	With kit lens	19999.00	https://images.unsplash.com/photo-1516035069371-29a1b244cc32?w=400&h=300&fit=crop
22	GoPro	GoPro Hero 9	Like new	Action cam	16000.00	https://images.unsplash.com/photo-1526170375885-4d8ecf77b99f?w=400&h=300&fit=crop
23	Amazon	Kindle Paperwhite	Good condition	8GB	4500.00	https://images.unsplash.com/photo-1507842217343-583bb7270b66?w=400&h=300&fit=crop
24	Sony	PlayStation 4	Good condition	500GB	14999.00	https://images.unsplash.com/photo-1607853202273-797f1c22a38e?w=400&h=300&fit=crop
25	boAt	boAt BassHeads 225	Like new	Wired	499.00	https://images.unsplash.com/photo-1484704849700-f032a568e944?w=400&h=300&fit=crop

```
MySQL localhost:33060+ ssl reloop_db SQL > select * from order_items;
```

id	order_id	product_id	quantity	price
2	17	13	1	1899.00
3	17	1	1	18999.00
4	18	11	1	55000.00
5	19	5	1	27000.00
6	20	14	1	2200.00
7	20	16	1	18000.00
8	21	10	1	30000.00
9	22	11	1	55000.00
10	22	18	1	1499.00
11	23	11	1	55000.00
12	24	7	1	7999.00
13	25	14	1	2200.00
14	25	13	1	1899.00

```
13 rows in set (0.0011 sec)
MySQL localhost:33060+ ssl reloop_db SQL >
```

localhost:5000/api/products

Pretty print


```
[
  {
    "id": 1,
    "brand": "Apple",
    "name": "iPhone 11",
    "condition_desc": "Good condition",
    "attributes": "64GB",
    "price": "18999.00",
    "image_url": "https://images.unsplash.com/photo-1574755393849-623942496936?w=400&h=300&fit=crop"
  },
  {
    "id": 2,
    "brand": "Apple",
    "name": "iPhone 12",
    "condition_desc": "Like new",
    "attributes": "128GB",
    "price": "24999.00",
    "image_url": "https://images.unsplash.com/photo-1605236453806-6ff36851218e?w=400&h=300&fit=crop"
  },
  {
    "id": 3,
    "brand": "Apple",
    "name": "iPhone 13",
    "condition_desc": "Good condition",
    "attributes": "128GB",
    "price": "34999.00",
    "image_url": "https://images.unsplash.com/photo-1632661674596-df8be070a5c5?w=400&h=300&fit=crop"
  },
  {
    "id": 4,
    "brand": "Samsung",
    "name": "Samsung M32",
    "condition_desc": "Like new",
    "attributes": "128GB",
    "price": "10499.00",
    "image_url": "https://images.unsplash.com/photo-1610945265064-0e34e5519bbf?w=400&h=300&fit=crop"
  },
  {
    "id": 5,
    "brand": "Samsung",
    "name": "Samsung S21",
    "condition_desc": "Good condition",
    "attributes": "256GB",
    "price": "27000.00",
    "image_url": "https://images.unsplash.com/photo-1616348436168-de43ad0db179?w=400&h=300&fit=crop"
  },
```